Chirp Watermark Detection Based on Fractional Auto-correlation Statistic

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Abstract. To overcome the shortage of the existing detector algorithm when the focus order of Chirp watermark near ± 1, a novel blind detection method based on fractional auto-correlation statistic has been proposed. The theory and methods of implementation have been discussed. Simulation results show that the algorithm is better robust to common attack.

Keywords: Fractional Fourier transform, digital watermarking, Chirp signal, fractional auto-correlation.

1 Introduction

The prompt booming of internet has brought abundant opportunities to commerce, scientific research and entertainment by the various services and means of electronic publishing and printing, electronic advertisement, digital library, internet videos and audios, e-business and etc. The digital multi-media products could be easily copied and extensively spread. Digital watermark technique provides a solution to this dilemma, which could be used for copy control and prohibiting unauthorized copy, and protects the copyright in transmission and application [1].

So far, several methods have been proposed to achieve this purpose. The digital watermark embedding and detecting algorithms mainly perform in the time/spatial domain or transform domain. Fractional Fourier transform shows the best focus property [2,3] on a given Chirp signal in a specific fractional order. The Chirp signal, with linear frequency and geometric symmetry and the relatively larger time-bandwidth product, represents relatively better robustness to common Stationary filtering and affine transform. Therefore two dimensional Chirp signal has been used as digital watermark embedded into image. The proposal of multi-component Chirp watermark algorithm borrowed the theory in [4] as to the combination of spatial domain/frequency domain technique, which directly embeds the Chirp watermark signal into the spatial domain of the image and the detects by Radon-Wigner transform (RWT) [3].

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Consider a watermarked image $I'(x,y)$ with the same size of $N \times N$ as the original image $I(x,y)$: $I'(x,y) = I(x,y) + \lambda C(x,y)$,Where λ is the watermark amplitude or strength, and a compromise between the robustness of the watermarking algorithm and the invisibility of the watermark. The 2-D multi-component and real-value chirp signal, used here as a watermark, has the following form:

$$C(x,y) = \sum_{n=1}^{K} A_n \cos\left(\pi\mu_{nx}x^2 + 2\pi f_{nx}x + \pi\mu_{ny}y^2 + 2\pi f_{ny}y\right) \tag{1}$$

Where K is the number of components and A_n is the amplitude of each component.

The RWT is equal to the squared magnitude of the FRFT of the signal [8], therefore a series of Chirp watermark detection methods based on fractional Fourier transform has been proposed recent years [5-7]. Although those detection methods could detect the Chirp watermark very well by virtue of concentration on Chirp signal, there are some limitations. Because the 2D fractional Fourier transform is the continuous time-frequency plane between spatial domain and frequency domain, when the transform order is approaching 1, the energy of the image is also in a process of converging to the center. Once the corresponding focus order of each Chirp watermark component approaches to ± 1, it would be inevitably influenced by the converging effect of image energy, which easily leads to difficulty of searching the Chirp concentration peak or leads to False alarm. Thus the watermark detection method proposed in [5-7] applies only to the situations in which the Chirp watermark focuses on a smaller FRFT order and the key space is restricted. For that reason, a novel blind detection method to2-D multi-component Chirp watermark has been proposed in this paper.

2 The Principles of Fractional Auto-Correlation Statistic and Chirp Signal Detection

As the ambiguity function (AF) of Chirp signal is a radial line through the origin of the ambiguity plane [2,3] and the slope *m* corresponds to the Chirp rate of the signal, integrating magnitude AF along a line with angle ϕ going through the origin equals to integrating fractional auto-correlation with the angle ϕ of the received signal, thus the following can be derived:

$$L(\phi) = \int \left|\left(s \otimes_{\phi} s\right)(\rho)\right| d\rho \tag{2}$$

Let Chirp rate perform as variable, the fractional auto-correlation statistic can be further rewritten as:

$$L(m) = \int \left|\left(s \otimes_{\arctan(m)} s\right)(\rho)\right| d\rho \tag{3}$$

The fractional auto-correlation of signals is given as:

$$\left(s \otimes_{\phi} s\right)(\rho) = F^{-\pi/2}\left\{\left|S^{\pi/2+\phi}(u)\right|^2\right\}(\rho) \tag{4}$$

The detection statistics derived from the fractional auto-correlation demands less computational loads than the calculation via AF when $M \ll N$. The multi-component Chirp signal in different level of noise can be well detected [8] using the fractional auto-correlation statistic detection as the peaks denote the Chirp rates respectively. And for non-Chirp signal, i.e. Gaussian white noise, the peak will not appear in this detector. Therefore, we believe that this method used to detect the Chirp watermark, especially in case of Chirp watermark detection with noise disturbance, has good potential.

3 2-D Multi-Component Chirp Watermark Detection Method

According to the above theoretical analysis, the detection based on the fractional auto-correlation statistic for two-dimensional multi-component Chirp watermark includes the following steps:

(1) According to the approximate range of the evaluated Chirp rate in x and y direction, select two groups of Chirp rate sample values μ_{xi} and μ_{yi} as followed, which correspond to the FRFT order when concentrating:

$$p_{xi} = \frac{\arctan \mu_{xi}}{\pi/2} \in \mathrm{P_x}, \quad p_{yi} = \frac{\arctan \mu_{yi}}{\pi/2} \in \mathrm{P_y} \tag{5}$$

(2) Calculate the FRFT with the order $(1+p_{xi}, 1+p_{yi})$ of the image $I'(x,y)$ to be detected using the fast FRFT in [9] : $F_{(1+p_{xi},1+p_{yi})}\left[I'(x,y)\right]$, $p_{xi} \in \mathrm{P_x}, p_{yi} \in \mathrm{P_y}$;

(3) Square the magnitude of the discrete FRFT result matrix, then find its inverse Fourier transform, which is described as:

$$\left[I' \otimes_{(p_{xi},p_{yi})} I'\right](u_x,u_y) = F_{(-1,-1)}\left\{\left|F_{(1+p_{xi},1+p_{yi})}\left[I'(x,y)\right]\right|^2\right\}(u_x,u_y) \tag{6}$$

(4) Calculate the detection statistic L as following:

$$L(p_{xi},p_{yi}) = \sum_{u_x}\sum_{u_y}\left[I' \otimes_{(p_{xi},p_{yi})} I'\right](u_x,u_y) \tag{7}$$

(5) Repeat the above operation with the other $p_{xi} \in \mathrm{P_x}, p_{yi} \in \mathrm{P_y}$ and obtain different detection statistic under these FRFT orders. Draw the curve of $L \sim (\mu_x, \mu_y)$ based on the correspondence between (p_x, p_y) and (μ_x, μ_y). Each peak position of statistic L in (μ_x, μ_y) plane in the figure corresponds to the Chirp rate of each component in the 2-D Chirp watermark signal.

4 Simulation and Analysis

For simplicity, in the simulation, the initial frequency and initial phase of Chirp watermark are set to zero, therefore the 2-D Chirp rate is used as the watermark key. The following parameters have already been dimensionless normalized.

A. Verification of the detection

The Chirp rate parameters of a four-component Chirp watermark signal are as followed: $\mu_{1x}=\mu_{1y}=0.1$, $\mu_{2x}=\mu_{2y}=0.18$, $\mu_{3x}=\mu_{3y}=0.3$, $\mu_{4x}=\mu_{4y}=0.45$. Embedding this watermark signal into a gray-scale image 'Lena' with 512×512 pixels according to Eq.(1) and Eq. (2). Consider the watermark amplitude λ=1.5.The PSNR between the watermarked image and the original image is 45.8dB.

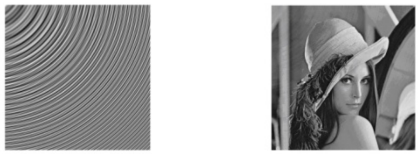

Fig. 1. Four-component Chirp watermark signal and the watermarked image

According to the given formula $p=2\mathrm{acot}(-\mu)/\pi$, the corresponding FRFT focus order of each component of the Chirp watermark is calculated as below: $p_{1x}=p_{1y}=-0.9365$, $p_{2x}=p_{2y}=-0.8866$, $p_{3x}=p_{3y}=-0.8145$, $p_{4x}=p_{4y}=-0.7308$. Calculate the fractional auto-correlation statistic *L*. Since the Chirp rates in x axis and y axis are equal to each other, we only search peaks along the diagonal in the (μ_x,μ_y) plane. Both x and y axis have 200 sampling points. As shown in Figure 2, the four peaks are very clear, of which the coordinates corresponding to Chirp rate parameters of the four components of the Chirp watermark signal. The simulation shows that when the focus order approaches to -1, this detection method still has a good result, which is not influenced by the image energy. Therefore, in this respect, the proposed method has explicit advantages than other existing FRFT-based Chirp watermark detection scheme[5-7].

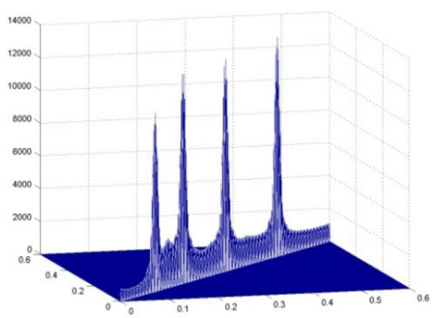

Fig. 2. The Chirp watermark detection based on fractional auto-correlation statistic

B. Robustness test simulation

In the following robustness test simulation, a four-component Chirp watermark signal with a large varying range of Chirp rate is embedded a into the image 'Lena' in the same way as the previous example. The Chirp rate parameter values after dimensional normalization are as followed: $\mu_{1x} = \mu_{1y} = 0.5$, $\mu_{2x} = \mu_{2y} = 1$, $\mu_{3x} = \mu_{3y} = 2$, $\mu_{4x} = \mu_{4y} = 4$. The PSNR between the watermarked image and the original image is 41.7dB.

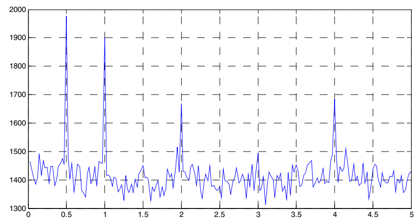

Fig. 4. The detector response when the watermarked image is corrupted with the additive white Gaussian noise (zero-mean, variance=20)

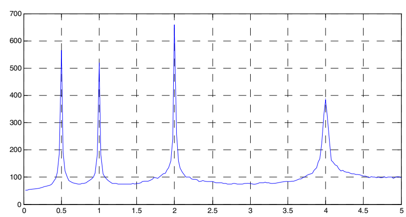

Fig. 5. The detector response when the watermarked image is cropped 50%

We detect the Chirp watermark in the attacked fig.1 which is once watermarked. The numbers of samples are both 200 in x-direction and y-direction. Noted in this case the searching range is relatively large, so the searching step length increases accordingly. Fig.4-5 show that when the image suffered certain attack, the Chirp watermark still can be robustly detected using methods described in this article. Neither the interval between chirp rates from various components of the Chirp watermark, nor the value range of the corresponding concentrating fractional orders, has effect on the test results, thus the program has a larger key space.

5 Conclusion

In this paper, a novel detection method of Chirp watermark with multi-component is proposed based on fractional auto-correlation statistic, which utilizes the relationship between the fractional correlation and ambiguity function of the Chirp signal. Simulations show that this blind detection method performs better when the corresponding FRFT focus order of Chirp rate approaches to ± 1 or the order of the components are close, and is robust to common attacks.

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